Flying DRONES Can Be Great Fun

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Drones, also known as unmanned aerial vehicles, were originally developed for military purposes. These are aerial vehicles without a pilot, which are controlled either from ground using radio controllers or are programmed for autonomous flights. Now, besides military action, drones are being built for various kinds of applications that are usually boring, routine or too dangerous for human beings.

Just like aero modelling, flying drones can be a great fun. However, their use is restricted by law in most countries. So before buying a drone, do check the law pertaining to use of drones in your region.

Fortunately, many small-size drones are now available in the market to experiment with. But because of their large variety and variance in size, it can be confusing to select the one most appropriate for you. Hope the hints below will be of some help to you.

Types of drones

Most drones are designed for specific purposes. There are drones that do anything from home delivery, photography, videography, racing or surveillance. Based on their functionality and build these can be divided into three main types:

1. Ready-to-fly drones that come fully equipped but require a clear fly zone and enough battery power to fly.
2. Bind and fly drones that do not come with a receiver. The compatible transmitter is brought in by you.
3. Almost ready-to-fly drones for DIY enthusiasts.

Phantom 4 drone by DJI

Propulsion

Drones meant for DIY enthusiasts generally have four or six propellers with separate motor for each. The motors and the propellers have to be exactly identical for a proper lift-off and movement in the air.

Bigger drones use brushless motors while the smaller ones use brush motors. These differ in architecture as brushless motors have spinning magnets instead of spinning coils. Brushless motors can deliver more torque and are longer lasting that the brushed ones. For a stable flight, everything including the motor and speed controller must work in unison.

What to look out for

Carbon fibre is one of the most desirable materials for the frame. It is lightweight and strong. Other materials are titanium, aluminium and steel. However, metals are heavy as compared to fibreglass and other polycarbonate materials. Apart from carbon fibre, Balsa wood frame is also light, tough and cost-effective.

Sensors. Almost all drones come with an accelerometer for measuring linear acceleration. It allows a radio-controlled drone to remain stable in the air with the help of a gyroscope that detects angular changes. There is also an inertia measurement unit, which is a small board that houses the gyroscope and accelerometer.

Higher-end drones also include a compass, barometer or GPS.

Battery life. Lithium-polymer rechargeable batteries power almost all drones. These are lightweight and pack a lot of power. These need to be charged regularly and also need to be handled cautiously.

Smartphone connectivity. Most commercial drones come with their proprietary remote controllers. Only some give you the freedom to connect to a smartphone in case you forget the controller at home. Make sure that your phone is compatible and syncs with the requirements for flying the drone in virtual mode. Some drones are also Wi-Fi-compatible and come with a setting to switch and select connectivity options within their respective mobile applications.